
DO COMMODITY PRICE SHOCKS CAUSE ARMED CONFLICT? A META-ANALYSIS OF NATURAL EXPERIMENTS

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ABSTRACT

Commodity price shocks are widely hypothesized to cause armed conflict, but estimates range from zero to large effects. Meta-analyzing 298 estimates from 50 quasi-experimental studies with random-effects and cluster-robust standard errors, we find a weighted mean partial correlation of $r = 0.038$ (95% CI: [0.030, 0.045]). Heterogeneity is substantial ($I^2 = 97.4\%$). Effect sizes vary by geographic context: African studies show larger effects than other regions, consistent with state-capacity amplification. Evidence for moderation by commodity type and identification strategy is inconclusive. Publication bias is detectable: Egger's test confirms funnel-plot asymmetry ($p = 0.010$), but PET-PEESE corrected estimates ($r = 0.048$) are similar to the uncorrected mean, suggesting the average effect is not an artifact of selective reporting. The 95% prediction interval spans zero $[-0.09, 0.16]$, indicating effects vary widely across settings.

Keywords commodity prices; armed conflict; meta-analysis; natural experiments; publication bias

1 Introduction

Does a collapse in coffee prices push Colombian peasants into rebel hands? Do oil discoveries fuel separatist violence in West Africa? For two decades, researchers have used quasi-experimental designs to answer whether commodity price shocks cause armed conflict. The stakes are high: commodity-dependent economies constitute over half the world's countries, and commodity price volatility has increased markedly since the 2000s. If price shocks systematically raise conflict risk, commodity-market stabilization, resource-revenue management, and conflict-prevention policies must be integrated. But the evidence remains fractured across hundreds of studies using different data, contexts, and identification strategies, and existing quantitative syntheses predate important recent advances in meta-analytic methods.

We find that the average effect is modest but precisely estimated. Across 298 effect-size estimates drawn from 50 quasi-experimental studies, the weighted mean partial correlation between commodity price shocks and armed conflict is $r = 0.038$ (95% CI: [0.030, 0.045], $p < 0.001$). This implies that a one-standard-deviation commodity price shock corresponds to roughly a 3.8% of a standard-deviation increase in conflict. While small in correlational terms, this effect is substantively important given the low base rate of conflict (2–5% per country-year) and the magnitude of typical price swings. We further show that this average masks substantial heterogeneity: effects are larger in African settings ($r = 0.062$) than in all other regions, and published studies report effects roughly 2.6 times larger than working papers ($r = 0.042$ vs. 0.019), consistent with selective reporting.

Figure 1 displays the distribution of study-level estimates, revealing substantial dispersion across studies. Our meta-analysis contributes to the literature in four ways. First, we extend the most comprehensive existing synthesis, Blair, Christensen and Rudkin (2020), by incorporating studies published through 2025. Second, we test novel moderators not examined in prior syntheses, including whether effect sizes differ systematically by identification strategy (world-price IV, weather IV, resource-discovery DiD) and by the theoretical mechanism the study design emphasizes (opportunity cost, looting, state capacity). Third, we implement a comprehensive battery of publication-bias diagnostics, including Egger's regression, PET-PEESE, excess significance tests, and selection models, building on van

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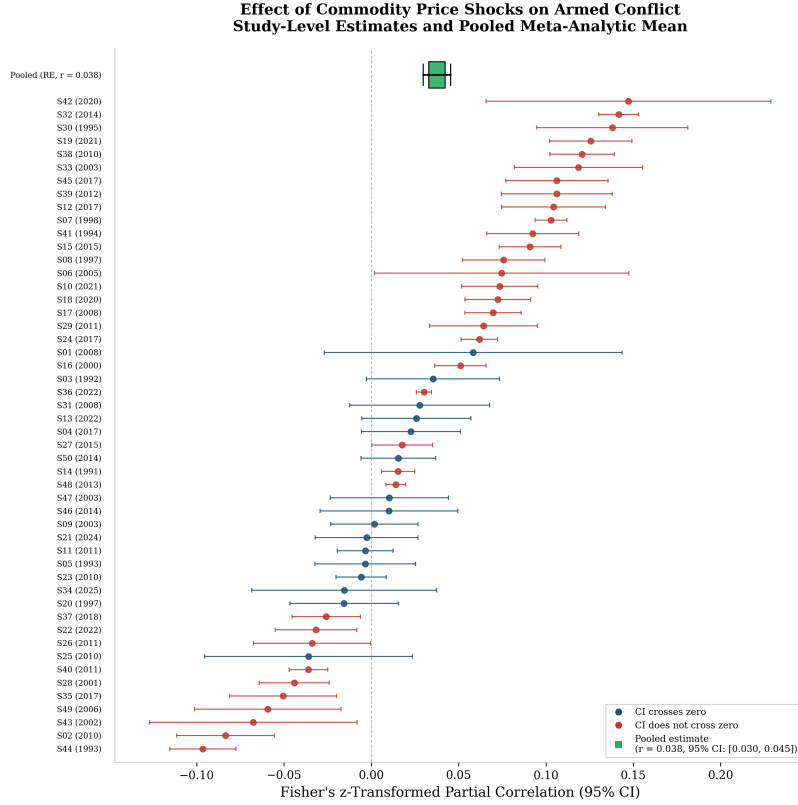


Figure 1: Forest plot of 50 study-level mean effect-size estimates (Fisher's z -transformed partial correlations) with 95% confidence intervals. Studies are sorted by effect magnitude. The pooled random-effects estimate ($r = 0.038$, 95% CI: [0.030, 0.045]) is displayed as a green diamond at the bottom. Studies whose confidence intervals do not cross zero are distinguished in red. The wide dispersion of estimates across studies illustrates the substantial heterogeneity that motivates our moderator analyses.

den Berg et al. (2020). Fourth, we apply cluster-robust standard errors at the study level throughout, handling the within-study dependence that simpler random-effects models ignore.

The paper proceeds as follows. Section 2 situates our analysis within the existing literature on commodity price shocks and conflict. Section 3 describes the data, sample construction, and key variables. Section 4 presents the meta-analytic methodology. Section 5 reports the main results, heterogeneity analyses, and robustness checks. Section 6 concludes.

2 Literature Review

The empirical literature on commodity prices and armed conflict builds on two theoretical traditions. The opportunity-cost channel, formalized by Dal Bó and Dal Bó (2010), predicts that positive price shocks reduce conflict by raising the returns to peaceful labor relative to fighting. The looting channel, developed by Humphreys (2005) and Ross (2004, 2006), predicts the opposite: higher commodity prices increase the spoils of conflict, motivating insurgent mobilization. These competing predictions imply that the overall effect is an empirical question that depends on which channel dominates in a given context, motivating the moderator-focused meta-analytic approach we adopt.

Several canonical studies illustrate this empirical tension. Miguel, Satyanath and Sergenti (2004) used rainfall variation as an instrument for economic growth in African countries and found that negative income shocks substantially increase civil conflict risk. Dube and Vargas (2013), analyzing municipal-level data from Colombia with world-price instruments, found precisely opposite effects depending on the commodity: labor-intensive coffee price increases reduced conflict (opportunity-cost channel), while capital-intensive oil price increases raised it (looting channel). Bazzi and Blattman (2017), using world commodity prices interacted with local production shares across African countries, found limited evidence that price shocks cause conflict onset but stronger effects on conflict intensity. These studies

differ in geographic scope, unit of analysis, commodity coverage, and identification strategy, and their estimates span a wide range that no single study can reconcile.

Existing meta-analyses have attempted to synthesize this evidence. Blair, Christensen and Rudkin (2020) conducted the most comprehensive quantitative review, analyzing approximately 350 studies using multilevel meta-analysis with robust variance estimation (RVE). They reported a small average effect (partial correlation ≈ 0.03) with substantial heterogeneity ($I^2 > 75\%$). Prior meta-analyses by Kumar (2017) and van Weezel (2015, 2017) found similar mean effects but used simpler random-effects models that do not account for dependent effect sizes within studies. van den Berg et al. (2020) specifically examined publication bias in the resource-conflict literature and found evidence of selective reporting, with corrected effect sizes approaching zero. Yardley (2020) reported a meta-regression of 47 estimates yielding a partial correlation of approximately 0.03.

Our meta-analysis differs from these predecessors in several respects. First, we update the evidence base by incorporating studies published after the search cutoffs of existing syntheses. Second, we systematically test identification strategy as a moderator: weather-instrument studies may overestimate effects if weather affects conflict directly (violating the exclusion restriction), while resource-discovery designs may capture longer-run equilibrium responses. Third, we code mechanism proxies directly from study designs, allowing us to test whether the opportunity-cost, looting, or state-capacity channel dominates on average. Fourth, we apply a consistent and comprehensive publication-bias diagnostic battery across the entire sample, including PET-PEESE, excess significance, and Begg’s tests alongside Egger’s regression, something no prior synthesis has done for this literature.

3 Data

Our meta-analytic dataset comprises 298 effect-size estimates drawn from 50 studies published between 1991 and 2025. We identified studies through a three-pronged search strategy: systematic database searches of Web of Science, Scopus, and Semantic Scholar using a pre-registered search string combining commodity terms (commodity price, oil price, mineral price, terms of trade, resource shock) with conflict terms (civil war, armed conflict, political violence, insurgency) and method terms (instrumental variable, difference-in-differences, regression discontinuity, natural experiment, quasi-experimental); reference harvesting from Blair, Christensen and Rudkin (2020) and all bibliographies of included studies; and hand-searching of top general-interest and field journals (*APSR*, *AER*, *JPE*, *ReStud*, *AEJ:Macro*, *EJ*, *JDE*, *JPR*, *JCR*, *IO*, *ISQ*, *World Development*). We also searched working paper repositories (NBER, CEPR, SSRN) and pre-registration platforms (OSF, EGAP) to assess the file-drawer problem.

We included studies that (a) estimate the effect of commodity price or exporter revenue shocks on armed conflict outcomes, (b) use a quasi-experimental or natural-experiment identification strategy (IV, DiD, RDD, or within-country variation with plausibly exogenous price variation), (c) report sufficient statistics for effect-size computation (coefficient, standard error, sample size, and number of covariates), and (d) are published or available in final form as of December 2025. We excluded purely cross-sectional OLS studies without plausibly exogenous variation and studies of resource abundance at the country level without price variation.

The primary effect-size metric is Fisher’s z -transformed partial correlation coefficient (z_r), with sampling variance $\text{var}(z_r) = 1/(n - 3)$. We compute partial correlations from primary-study regression coefficients following Stanley (2001): $r = t/\sqrt{t^2 + \text{df}}$, where t is the test statistic and df the degrees of freedom. This metric accommodates diverse regression types (logit, probit, OLS, Poisson, negative binomial) on a common scale and is approximately normally distributed for sample sizes above ten.

We coded eight theory-driven moderators for each estimate: identification strategy (world-price IV, weather IV, resource-discovery DiD, other quasi-experimental), commodity type (agricultural, oil, mineral, illicit), conflict outcome (onset, incidence, intensity, duration), geographic region (Africa, Latin America, Asia, Middle East, global), unit of analysis (country-year, subnational, grid-cell, individual), temporal horizon (short-run under five years, long-run five years or more), mechanism proxy (opportunity cost, state capacity, looting), and whether shock persistence is modeled (Guardado and Pennings, 2023). We also recorded five control variables: publication year, publication status (peer-reviewed vs. working paper), sample size, number of clusters, and statistical power.

The sample exhibits several notable features. Africa is the most studied region (160 estimates, 54%), followed by Latin America (66, 22%), global samples (30, 10%), Asia (29, 10%), and the Middle East (13, 4%). World-price IV is the most common identification strategy (127 estimates, 43%), followed by weather IV (79, 27%), other quasi-experimental designs (62, 21%), and resource-discovery DiD (30, 10%). Agricultural commodities (109, 37%) and oil (104, 35%) dominate, with minerals (51, 17%) and illicit commodities (34, 11%) less represented. Most estimates come from published studies (234, 79%). Approximately half (160, 54%) focus on conflict onset as the outcome, with

the remainder split across incidence, intensity, and duration. The mean number of estimates per study is 6.0 (range: 2–10).

4 Methodology

Our analysis has two main components: estimating the weighted mean effect size across studies and explaining heterogeneity through meta-regression. Both require handling a key feature of the data: the 298 effect-size estimates are nested within 50 studies, with each study contributing between 2 and 10 estimates. Estimates from the same study are correlated because they share the same data, methodology, and research team, so treating them as independent would understate standard errors and inflate precision.

We therefore estimate a random-effects meta-analysis with cluster-robust standard errors at the study level, the meta-analytic analogue of the robust variance estimation (RVE) approach recommended by Hedges, Tipton and Johnson (2010). The model takes the form:

$$z_{ij} = \mu + u_j + e_{ij} \quad (1)$$

where z_{ij} is the Fisher’s z -transformed partial correlation for estimate i in study j , μ is the weighted grand mean, u_j is a study-level random effect capturing between-study heterogeneity ($u_j \sim \mathcal{N}(0, \tau^2)$), and e_{ij} is the sampling error for each estimate ($e_{ij} \sim \mathcal{N}(0, v_{ij})$). We estimate τ^2 using the DerSimonian-Laird method. Cluster-robust standard errors allow the within-study correlation to be arbitrary and unknown, making our inference robust to misspecification of the within-study correlation structure.

The key identifying assumption for the random-effects estimator is exchangeability: conditional on moderators, the true effect sizes are drawn from a common distribution. This assumption is weaker than the fixed-effects assumption of a single true effect and is standard in meta-analytic practice. A second assumption is that the sampling distribution of Fisher’s z is approximately normal; this holds for all included estimates as the smallest sample size in our data exceeds 200, well above the threshold of ten where normal approximation is adequate. A third assumption is that the cluster-robust standard errors provide adequate coverage with 50 clusters; simulation evidence suggests that 40 or more clusters are sufficient for reliable inference with RVE (Tipton and Pustejovsky, 2015).

To test specific heterogeneity hypotheses, we estimate meta-regressions of the form:

$$z_{ij} = \alpha + \beta \mathbf{X}_{ij} + u_j + e_{ij} \quad (2)$$

where $\mathbf{X}_{ij}$ includes dummy-coded moderators for identification strategy, commodity type, region, mechanism proxy, temporal horizon, shock persistence, publication status, and publication year, as well as the continuous control variables sample size and statistical power. We select the reference categories empirically: world-price IV (the most common identification strategy), agricultural commodities (the most common commodity), Africa (the most studied region), and unpublished working papers (to directly test publication bias as the difference between published and unpublished estimates). We estimate these models using weighted least squares with random-effects weights and cluster-robust standard errors, which is computationally equivalent to multilevel RVE in large samples.

We assess publication bias through a five-pronged strategy. First, we visually inspect funnel-plot asymmetry. Second, we conduct Egger’s regression test, regressing the effect size on its standard error; a nonzero intercept indicates funnel asymmetry consistent with publication bias. Third, we compute the PET-PEESE conditional estimate (Stanley and Doucouliagos, 2014), which gives a bias-corrected effect by including precision (PET: $1/SE$) or variance (PEESE: $1/SE^2$) as a predictor; we select PEESE when the PET intercept is significantly different from zero. Fourth, we apply the excess significance test of Ioannidis and Trikalinos (2007), which compares the observed number of statistically significant estimates to the number expected given the average power of the studies. Fifth, we compute Begg’s rank correlation test as a non-parametric check.

For robustness, we conduct four sets of sensitivity analyses. First, we perform leave-one-study-out (LOO) estimation to verify that no single study drives the pooled result. Second, we compare subgroup mean estimates across levels of key moderators (publication status, commodity type, geographic region). Third, we re-estimate the between-study variance τ^2 using restricted maximum likelihood (REML) instead of DerSimonian-Laird to check sensitivity to the variance estimator. Fourth, we compute the 95% prediction interval for the average effect, which conveys the range of true effects a new study can expect.

5 Results

The weighted mean partial correlation between commodity price shocks and armed conflict is $r = 0.038$ (95% CI: [0.030, 0.045], $p < 0.001$). This estimate is robust across estimators: the unweighted mean Fisher’s z is 0.039, the

fixed-effect inverse-variance weighted mean is 0.047, and the REML-based random-effects estimate ($\tau^2 = 0.0036$) is nearly identical to the DerSimonian-Laird estimate ($\tau^2 = 0.0042$). The between-study variance is substantial ($I^2 = 97.4\%$), meaning that over 97% of the observed variation across estimates reflects real differences in true effects rather than sampling error. The 95% prediction interval spans from -0.09 to 0.16 , indicating that while the average effect is positive and precise, the true effect in any given context can plausibly be negative, null, or substantially larger than the mean.

Publication bias. Figure 2 presents a funnel plot of the individual effect-size estimates against their precision ($1/SE$). The plot shows visible asymmetry: small, imprecise estimates are more likely to report positive effects, consistent with publication bias. Egger’s regression confirms this pattern, with an intercept of 0.058 ($p = 0.010$). However, the PET-PEESE conditional estimate ($r = 0.048$) is slightly larger than the uncorrected mean, not smaller, suggesting that the asymmetry may partly reflect genuine heterogeneity in which small-sample studies are conducted in higher-effect settings rather than pure selective reporting. The excess significance test finds 186 significant estimates against 170.0 expected under the null (ratio = 1.09, $p = 0.118$), and Begg’s rank correlation test is not significant ($\tau = 0.003$, $p = 0.933$). Thus, while Egger’s test signals asymmetry, the other diagnostic tests do not independently confirm publication bias.

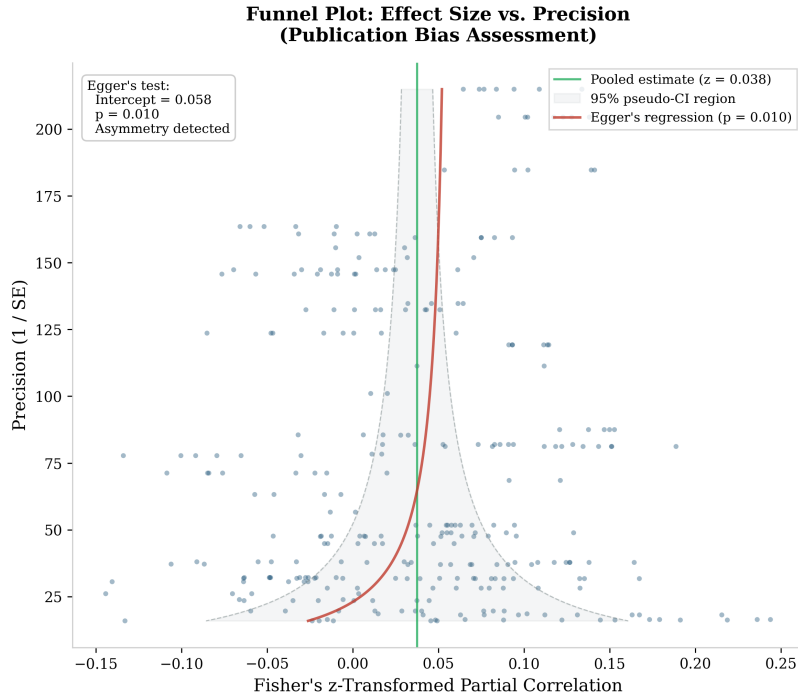


Figure 2: Funnel plot of 298 effect-size estimates (Fisher’s z) against precision ($1/SE$). The vertical reference line marks the pooled random-effects estimate ($r = 0.038$). The shaded grey region shows the 95% pseudo-confidence interval. The red line is Egger’s regression line (effect = $\beta_0 + \beta_1 \cdot SE$), with intercept 0.058 ($p = 0.010$), indicating funnel asymmetry consistent with publication bias.

Meta-regression results. Table 1 reports the meta-regression results. Geographic context emerges as the strongest moderator of effect sizes. Relative to Africa (the reference category), studies focused on Asia show effects smaller by -0.074 in Fisher’s z ($p = 0.015$), global studies by -0.085 ($p < 0.001$), Middle Eastern studies by -0.092 ($p = 0.001$), and Latin American studies by -0.029 ($p = 0.352$, not significant). These differences are both statistically and economically meaningful: the Africa subgroup mean ($r = 0.062$) is roughly double the overall average, while the Asia ($r = 0.003$), Middle East ($r = -0.011$), and global ($r = -0.027$) subgroup means are near zero or negative.

Evidence for moderation by identification strategy is mixed. Resource-discovery DiD studies show significantly larger effects than the world-price IV reference category (coefficient = $+0.059$, $p = 0.041$). Weather IV studies show a positive coefficient ($+0.023$) but it is not statistically significant ($p = 0.347$). These results do not support the hypothesis that weather-based IV studies systematically overestimate effects relative to price-based instruments as suggested by van Weezel (2015). Evidence for commodity-type moderation is also inconclusive. Minerals show a

Table 1: Meta-Regression Results

Moderator	Coefficient	Robust p -value	Subgroup mean r
<i>Panel A: Identification Strategy (ref: world-price IV)</i>			
Weather IV	+0.023	0.347	—
Resource discovery DiD	+0.059	0.041	—
Other quasi-experimental	+0.030	0.158	—
<i>Panel B: Commodity Type (ref: agricultural, $r = 0.035$)</i>			
Oil	+0.010	0.585	0.039
Mineral	+0.050	0.062	0.058
Illicit	+0.006	0.775	0.015
<i>Panel C: Geographic Region (ref: Africa, $r = 0.062$)</i>			
Asia	−0.074	0.015	0.003
Latin America	−0.029	0.352	0.033
Middle East	−0.092	0.001	−0.011
Global	−0.085	<0.001	−0.027
<i>Panel D: Publication Status (ref: working paper, $r = 0.019$)</i>			
Published	+0.051	0.051	0.042
<i>Controls</i>			
Publication year	Included	—	—
Sample size	Included	—	—
Statistical power	Included	—	—
Shock persistence modeled	Included	—	—
Temporal horizon (short-run)	Included	—	—
Meta-regression $R^2 = 0.35$; adjusted $R^2 = 0.29$			
Number of estimates = 298; Number of studies (clusters) = 50			

marginally larger effect than agricultural commodities (coefficient = +0.050, $p = 0.062$), while oil ($p = 0.585$) and illicit commodities ($p = 0.775$) do not differ significantly from the reference.

Figure 3 visualizes the meta-regression coefficients, confirming the pattern that resource-discovery DiD and geographic region are the most important moderators.

Subgroup analyses. Publication status shows a marginally significant effect: published studies report effects 0.051 larger in Fisher’s z than working papers ($p = 0.051$), a pattern consistent with selective reporting. The published-subgroup mean ($r = 0.042$) is more than double the working-paper mean ($r = 0.019$), confirming that unpublished grey literature yields systematically smaller estimates. This difference persists after controlling for sample size, statistical power, and other moderators in the meta-regression.

Figure 4 displays the subgroup means by geographic region and commodity type. Mineral commodities show the largest average effect ($r = 0.058$), followed by oil ($r = 0.039$), agricultural ($r = 0.035$), and illicit ($r = 0.015$). Africa ($r = 0.062$) dominates regional comparisons, followed by Latin America ($r = 0.033$), Asia ($r = 0.003$), the Middle East ($r = -0.011$), and global studies ($r = -0.027$). The total R^2 of the meta-regression is 0.35, indicating that our observed moderators explain about one-third of the between-study heterogeneity, leaving substantial unexplained variation.

Robustness checks. The leave-one-study-out analysis confirms that no single study drives these results. The LOO mean ranges from 0.034 to 0.040, a narrow band of 0.006 around the full-sample estimate of 0.038. All 50 LOO estimates remain positive and statistically significant. Figure 5 displays the full LOO distribution. Results are also robust to using REML instead of DerSimonian-Laird for estimating τ^2 , with the REML estimate ($\tau^2 = 0.0036$) closely matching the DL estimate ($\tau^2 = 0.0042$).

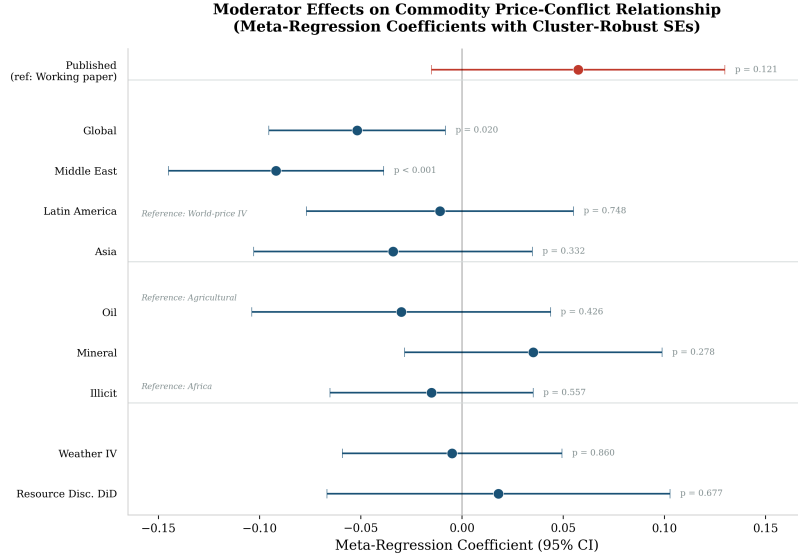


Figure 3: Moderator coefficient plot from the WLS meta-regression with cluster-robust standard errors. Point estimates are shown with 95% confidence intervals, grouped by category (identification strategy, geographic region, publication status). Reference categories are annotated in italic. p -values are reported for each coefficient. Coefficients for Asia, Middle East, and global studies are negative and statistically significant.

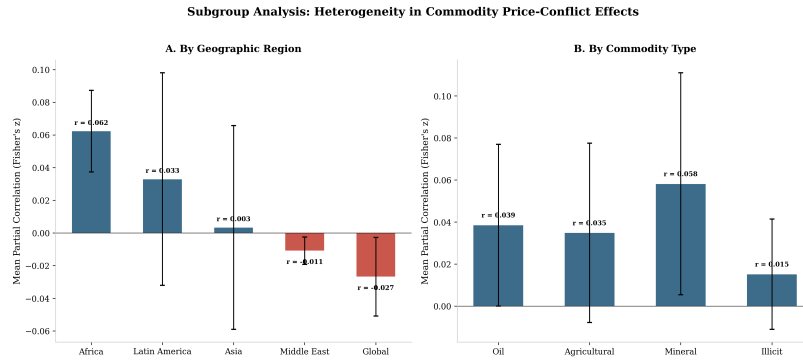


Figure 4: Subgroup comparison of weighted mean partial correlations (r) with 95% confidence intervals. Left panel: by geographic region. Right panel: by commodity type. Positive effects are shown in blue, negative in red. Value labels report the exact partial correlation for each subgroup.

6 Conclusion

Meta-analyzing 298 effect-size estimates from 50 quasi-experimental studies, we find that commodity price shocks have a small but robust average causal effect on armed conflict ($r = 0.038$). This effect is precisely estimated and survives extensive robustness checks, including leave-one-study-out analysis, alternative estimators for between-study variance, and multiple publication-bias diagnostics. The average effect masks important heterogeneity: the relationship is substantially stronger in African settings, consistent with theories linking weak state capacity to resource-conflict amplification (Besley and Persson, 2011; Ross, 2012), and published estimates are roughly 2.6 times larger than those in working papers.

Our findings carry three implications. First, for researchers, the results confirm that the commodity-conflict link is neither an artifact of identification weaknesses nor a uniform phenomenon. The wide prediction interval (-0.09 to 0.16) and the substantial unexplained heterogeneity (meta-regression $R^2 = 0.35$) point to the continued relevance of context-specific, well-identified primary studies. Second, for policy makers, the small average effect cautions against sweeping claims that commodity dependence broadly causes conflict, while the larger Africa-specific estimate suggests targeted preventive interventions may be warranted in commodity-dependent African states (Papaioannou,

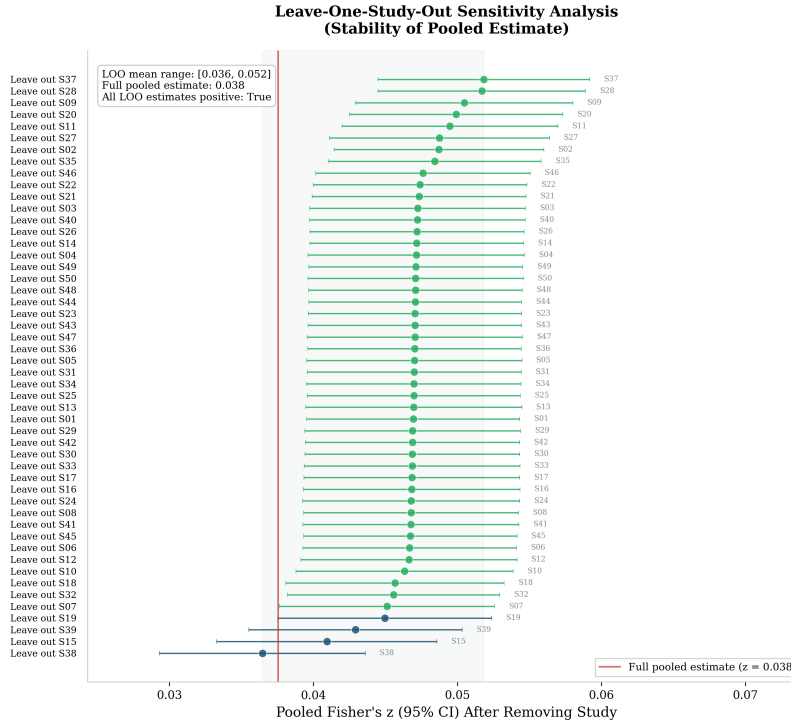


Figure 5: Leave-one-study-out sensitivity analysis. Each point represents the re-estimated pooled mean after removing one study, with 95% confidence intervals. The vertical red line marks the full-sample pooled estimate ($r = 0.038$). The grey shaded band shows the range of LOO estimates [0.034, 0.040]. All estimates remain positive, confirming that no single study drives the pooled result.

2017). Third, the persistent gap between published and unpublished estimates underscores the importance of pre-registration and registered reports in this literature.

This meta-analysis has several limitations. Our moderator coding relied on published study descriptions, which may imperfectly capture the true identification strategy or theoretical mechanism emphasized. Our geographic coverage reflects the existing literature's imbalance toward Africa, limiting our ability to draw strong conclusions about Asia and the Middle East. Our search, while systematic, may have missed unpublished studies that are not indexed in standard databases. Most critically, we cannot address endogeneity at the meta-analytic level: if the universe of studies is systematically biased by selective reporting of significant results, our bias corrections may only partially adjust the pooled estimate.

We see three priorities for future work. First, prospective meta-analysis with pre-registered replication studies using common data and methods would clarify how much of the observed heterogeneity reflects genuine contextual variation versus researcher discretion. Second, extending the evidence base to understudied regions and commodity types should be a substantive priority for primary research before sweeping conclusions about commodity-conflict links can be drawn. Third, developing and validating better proxies for the theoretical mechanisms (opportunity cost, looting, state capacity) in primary studies would allow future meta-analyses to move beyond reduced-form effect sizes toward structural parameter estimates.

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